

Quantum / Effective Action:

$$\Gamma(\varphi) \equiv -\frac{1}{2} \int \frac{d^d k}{(2\pi)^d} \tilde{\varphi}(-k) (k^2 + m^2 - \Pi(k^2)) \tilde{\varphi}(k) \\ + \sum_{n=3}^{\infty} \frac{1}{n!} \int \frac{d^d k_1}{(2\pi)^d} \dots \frac{d^d k_n}{(2\pi)^d} (2\pi)^d \delta^d(k_1 + \dots + k_n) \times V_n(k_1, \dots, k_n) \tilde{\varphi}(k_1) \dots \tilde{\varphi}(k_n)$$

$$\tilde{\varphi}(k) = \int d^d x e^{-ikx} \varphi(x)$$

- The tree-level Feynman diagrams it generates gives the Complete scattering amp. of the original theory

$$Z[J] = \int D\varphi \exp[iS(\varphi) + i \int d^d x J \varphi]$$

and

$$Z(J) = e^{iW[J]}$$

* Consider now

$$Z_{\Gamma}(J) \equiv \int D\varphi \exp[i\Gamma(\varphi) + i \int d^d x J \varphi] = e^{iW_{\Gamma}(J)}$$

i.e. we replace $S(\varphi)$ with the Quantum action $\Gamma(\varphi)$. $W_{\Gamma}(J)$ is given by the sum of connected diagrams (with sources) in which each line represents the exact prop Δ and each n -pt vertex rep the exact $1PI$ vertex V_n .

- Isolate the tree-level contribution to a path integral by introducing a dimensionless parameter t

$$Z_{\Gamma,t}(J) \equiv \int D\varphi \exp\left[\frac{i}{t} \left(\Gamma(\varphi) + \int d^d x J \varphi\right)\right] \\ = \exp[iW_{\Gamma,t}(J)]$$

- Every $\Delta \rightarrow t \Delta$
 $J \rightarrow \frac{1}{t} J$
 $V_n \rightarrow \frac{1}{t} V_n$

t P-E-V

P : # of propagators

$$P - E - V = L - 2$$

* $W[\mathcal{J}] = \sum_{L=0}^{\infty} \frac{1}{h^{L-1}} W_{L,L}[\mathcal{J}]$, limit $h \rightarrow 0$
 $L=0$: tree-level

$$W[\mathcal{J}] = W_{L,L=0}[\mathcal{J}]$$

* Stationary Phase Method:

$$Z[\mathcal{J}] = \int D\varphi \exp\left(\frac{i}{h} (S(\varphi) + \mathcal{J}\varphi)\right)$$

• Main Contribution comes from the critical point φ^i where $(S(\varphi) + \mathcal{J}\varphi)$ is stationary

• Suppose $\mathcal{J}=0$: $\int d\varphi e^{\frac{i}{h} S(\varphi)}$
 the largest contribution would come from the minimum of $S(\varphi)$: it would be the "slowest" to be suppressed as $h \rightarrow 0$
 i.e.

$$S(\varphi = \varphi^i) = S(\varphi^i) + \partial_{\varphi} S(\varphi^i) \varphi^i + S''(\varphi^i) \varphi^i{}^2 + \dots$$

* Now, we find the minimum of

$$S(\varphi) + \int d^d x \mathcal{J}(x) \varphi(x)$$

$$\frac{\partial S(\varphi)}{\partial \varphi(x)} + \mathcal{J}(x) = 0$$

$$\boxed{\frac{\partial S}{\partial \varphi_i} = -\mathcal{J}_i}$$

$$\varphi_0 = \varphi_0(\mathcal{J})$$

• Assume $\exists$ only a finite # of points

$$\alpha = 1, \dots, p. \quad \varphi_{0,\alpha}$$

All of which are isolated in a neighborhood

$$M_{\alpha} \text{ s.t. } M_{\alpha_i} \cap M_{\alpha_j} = \emptyset \quad \forall i, j$$

... let's look at the 2D case

$$a < x_0 < b \quad \text{s.t.} \quad S'(x_0) = 0, \quad S''(x_0) \neq 0$$

$$x_0: \text{maximum} \Rightarrow S''(x_0) < 0$$

$$S(x) = S(x_0) + S''(x_0) \frac{(x-x_0)^2}{2} + \mathcal{O}((x-x_0)^3)$$

$$F(\lambda) = \int_a^b \varphi(x) e^{\lambda S(x)} dx$$

$$= \left(\int_a^{x_0-\varepsilon} \int_{x_0-\varepsilon}^{x_0+\varepsilon} \int_{x_0+\varepsilon}^b \right) \varphi(x) e^{\lambda \left(S(x_0) + S''(x_0) \frac{(x-x_0)^2}{2} + \mathcal{O}((x-x_0)^3) \right)} dx$$

$$= \int_a^b dx \varphi(x) e^{\lambda S(x_0)} e^{\lambda S''(x_0) \frac{(x-x_0)^2}{2} + \mathcal{O}((x-x_0)^3)}$$

$$\varphi(x) = \left(\varphi(x_0) + \sum_{n=1}^{\infty} \frac{\varphi^{(n)}(x_0)}{n!} (x-x_0)^n \right)$$

$$e^{\lambda S(x_0)} \int_a^b dx \left(\varphi(x_0) + \sum_{n=1}^{\infty} \frac{\varphi^{(n)}(x_0)}{n!} (x-x_0)^n \right) e^{\lambda S''(x_0) \frac{(x-x_0)^2}{2}}$$

$$= e^{\lambda S(x_0)} \int_a^b dx \varphi(x_0) e^{\lambda S''(x_0) \frac{(x-x_0)^2}{2}} \left(1 + \mathcal{O}((x-x_0)^2) \right)$$

$$a \rightarrow -\infty, \quad b \rightarrow \infty$$

$$\sim e^{\lambda S(x_0)} \varphi(x_0) \sqrt{\frac{2\pi}{-S''(x_0) \lambda}}$$

$$\sim e^{\lambda S(x_0)} \lambda^{-1/2} \varphi(x_0) \sqrt{\frac{2\pi}{-S''(x_0)}}$$

Going back to QFT:
Quantum E.O.M:

$$\frac{\delta \Gamma(\varphi)}{\delta \varphi(x)} = -J(x)$$

Solution to this equation is

$$\varphi_J(x)$$

$$Z[h[J]] \sim e^{\frac{i}{\hbar} (\Gamma(\varphi_J) + \int d^d x J \varphi_J) + \mathcal{O}(\hbar^0)}$$

$$W[J] = \Gamma(\varphi_J) + \int d^d x J \varphi_J$$

$$* \langle \Omega | \phi(x) | \Omega \rangle = \frac{\delta}{\delta J(x)} W[J] \Big|_{J=0}$$

for $J \neq 0$

$$\langle \Omega | \phi(x) | \Omega \rangle_J = \frac{\delta}{\delta J(x)} W[J]$$

$$\langle \Omega | \phi(x) | \Omega \rangle_J = \frac{\delta}{\delta J(x)} \left(\Gamma(\varphi_J) + \int d^d x J \varphi_J \right)$$

Recall that $\varphi_J = \varphi(x, J)$ and so

$$\langle \Omega | \phi(x) | \Omega \rangle_J = \frac{\delta \Gamma(\varphi_J)}{\delta J(x)} + \varphi_J(x) + \int d^d y J(y) \frac{\delta \varphi_J(y)}{\delta J(x)}$$

• Use the chain rule on $\frac{\delta \Gamma}{\delta J}$:

$$\frac{\delta \Gamma(\varphi_J)}{\delta J(x)} = \int d^d y \frac{\delta \Gamma(\varphi_J)}{\delta \varphi_J(y)} \frac{\delta \varphi_J(y)}{\delta J(x)}$$

$$\langle \Omega | \phi(x) | \Omega \rangle_J = \int d^d y \left(\frac{\delta \Gamma(\varphi_J)}{\delta \varphi_J(y)} + J(y) \right) \frac{\delta \varphi_J(y)}{\delta J(x)} + \varphi_J(x)$$

• However, from the Quantum E.O.M we find variables

$$\langle \Omega | \phi(x) | \Omega \rangle_J = \varphi_J(x)$$